

“Math didn’t fail you — the way we talk about math failed you.”

Core Thesis

The barrier between “everyday people” and “advanced mathematics” is largely **linguistic and psychological**, not conceptual. Most people already understand the *ideas* behind many intimidating mathematical concepts — they just don’t recognize them because of jargon, notation, and cultural fear.

I. The Statistics — Math Anxiety Is Widespread (and Misplaced)

- **64% of Americans** report experiencing math anxiety, with **13%** describing it as severe. [\[1\]](#) [\[2\]](#)
- Women report higher rates (**70%**) compared to men (**57%**).
- **47%** believe math anxiety has set them back academically or professionally.
- Yet the APA notes that people with high math anxiety **often know how to do the math** — it’s an emotional reaction that interferes, not a lack of skill. [\[3\]](#)

Key takeaway for the speech: The majority of people who say “I’m bad at math” are actually saying “math *language* scares me.”

II. The Key Research — It’s a Language Barrier, Not a Conceptual One

Sheila Tobias — *Overcoming Math Anxiety* (1978)

The foundational work on this topic. Tobias found:

“The barrier to math is often linguistic, not conceptual. Many students can handle the mathematical concepts but become unnerved by the unfamiliarity of the symbols, terminology, and structure.”

Anna Sfard — *Thinking as Communicating* (2008)

Describes mathematics as a **language/discourse**. People “hit a wall” not because they can’t *do* the math, but because they can’t *parse* the formal language it’s written in.

Ashcraft & Krause (2007) — Working Memory & Math Anxiety

Decoding unfamiliar notation consumes **cognitive resources** (working memory) that would otherwise be used for actual problem-solving, creating a vicious cycle of anxiety and failure.

Finlayson, Maureen (2014) — “Addressing Math Anxiety in the Classroom”

Found that introducing concepts via **plain language first**, then moving to formal notation, significantly boosts confidence and performance.

III. The Killer Examples — “Scary” Math That’s Actually Simple

This is the heart of the speech. Walk the audience through these:

Intimidating Term	What It Actually Means	You Already Know This Because...
Irreducible Quintic / Polynomial	An equation with x raised to powers (like $x^5 + 3x^2 - 7 = 0$) that can't be simplified further	You've solved "what number times itself equals 9?" — that's a polynomial ($x^2 = 9$)
Ansatz	An educated guess you use as a starting point to solve a problem	You do this every time you estimate a tip, guess how long a drive will take, or eyeball whether furniture will fit in a room
Number Theory (Modular Arithmetic)	The study of whole numbers and their relationships — including "clock math," where numbers wrap around after reaching a value	You do modular arithmetic every time you read a clock. "It's 10am and the meeting is in 5 hours" → $10 + 5 = 15$, but on a 12-hour clock that's 3 o'clock . That's mod 12.
Abstract Algebra (Group Theory)	A "group" is a set of actions you can perform and undo, following specific rules	Every way you can rotate your phone and have the screen still look right forms a mathematical group. Every sequence of Rubik's Cube moves is group theory in your hands.
Real Analysis (Epsilon-Delta Limits)	A rigorous way of saying "you can get <i>as close as you want</i> to a target value"	Your thermostat. You want the room at 72°F. No matter how precise your comfort demand (within 0.5°, 0.1°, 0.01°...), you can always adjust the dial to hit that window. That "challenge and response" game <i>is</i> epsilon-delta.
Euclidean Geometry	The geometry of flat surfaces — points, lines, angles, triangles, circles	Every time you measure a room with a tape measure, check that a picture frame is level, or know that a triangle's angles add up to 180°, you're doing Euclidean geometry. You've been doing it since you drew shapes in grade school.

Intimidating Term	What It Actually Means	You Already Know This Because...
Differential Geometry (Geodesics)	The study of how curved surfaces bend and twist, and what “straight lines” look like on them	When you fly from New York to London, the plane doesn’t follow a straight line on the map — it curves north over the Atlantic. That curved “great circle” path is the shortest route on a sphere. That’s a geodesic , the core object of differential geometry. Your GPS solves differential geometry every time it gives you directions on a round Earth.
Diffeomorphism	A smooth, reversible transformation — reshaping something with no tearing, cutting, or gluing	Stretching pizza dough. You pull and reshape it, but you never tear it or glue pieces together, and you <i>could</i> smoosh it back to its original shape. That smooth, reversible deformation is literally a diffeomorphism.
Combinatorics	The math of counting arrangements and possibilities	“I have 4 shirts and 3 pants — how many outfits can I make?” ($4 \times 3 = 12$). You solved a combinatorics problem. Choosing a pizza with 3 toppings from a menu of 10? That’s “n choose k” — the heart of combinatorics.
Graph Theory	The study of networks — dots (vertices) connected by lines (edges)	Your social network. Each person is a dot, each friendship is a line. “How many people do I need to go through to reach Beyoncé?” is the shortest path problem — the foundational question of graph theory. Google Maps routing, airline connections, and even your LinkedIn “2nd degree connections” are all graph theory.
Derivative	How fast something is changing <i>right now</i>	Your car’s speedometer IS a derivative — it shows the rate of change of your position at this exact instant
Integral	Adding up a bunch of tiny pieces to get a total	Estimating total gas cost on a road trip where prices vary along the way — you’re mentally summing up small chunks, which is integration

Intimidating Term	What It Actually Means	You Already Know This Because...
Eigenvalue / Eigenvector	A direction that doesn't change when you transform something, just gets stretched or squished	Pulling a rubber band — the direction along the band that just stretches (doesn't rotate) is the eigenvector; how much it stretches is the eigenvalue
Fourier Transform	Breaking a complex signal into simple waves	Hearing a musical chord and picking out the individual notes — your ear is performing a Fourier transform
Topology	The study of shapes that can stretch but not tear	You intuitively know a donut and a coffee mug with one handle are “the same shape” — you could mold one into the other without cutting. That equivalence <i>is</i> topology.
Asymptote	A line a curve approaches but never quite reaches	“Almost there but never quite” — like Zeno's paradox of always halving the remaining distance to a wall
Optimization	Finding the best option given constraints	Choosing the fastest route on Google Maps, fitting everything into a suitcase, or maximizing your grocery haul on a budget — all optimization

IV. Math People Already Use Daily (Without Realizing It)

This connects the abstract to the concrete:

- **Proportional reasoning** — doubling a recipe, calculating sale discounts
- **Spatial geometry** — parking a car, packing a suitcase, rearranging furniture
- **Probabilistic thinking** — deciding whether to bring an umbrella, assessing risk
- **Logical deduction** — process of elimination (who ate the last cookie?)
- **Pattern recognition** — predicting the next song an algorithm will suggest
- **Budgeting arithmetic** — comparing unit prices at the grocery store

V. Persuasive Argument Structure (Suggested)

1. **Hook:** Ask the audience “How many of you would say you’re *bad* at math?” (expect ~60%+ of hands). Then say: “By the end of this speech, I’ll prove that every one of you already understands calculus, probability, and topology.”
 2. **Problem:** 64% of Americans have math anxiety. This costs people careers, money (37% say it hinders financial decisions), and confidence.
 3. **Cause:** The real barrier is **language**, not ability. Cite Tobias: “*The barrier to math is often linguistic, not conceptual.*”
 4. **Proof:** Walk through 3-4 killer examples from the table above. Show the scary term → reveal the simple idea → show them they already do it.
 5. **Implications:** If math is a language problem, it’s *solvable*. We can teach math differently — plain language first, formal notation second (cite Finlayson 2014).
 6. **Call to Action:** Stop saying “I’m bad at math.” You’re not. You just haven’t learned the dialect yet. And we should demand that math be taught in a way that builds on what people *already* understand.
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VI. Works Cited / Further Reading

- Tobias, Sheila. *Overcoming Math Anxiety*. Norton, 1978.
- Sfard, Anna. *Thinking as Communicating: Human Development, the Growth of Discourses, and Mathematizing*. Cambridge UP, 2008.
- Ashcraft, M. H., & Krause, J. A. “Working memory, math performance, and math anxiety.” *Current Directions in Psychological Science*, 16(4), 2007.
- Finlayson, Maureen. “Addressing Math Anxiety in the Classroom.” *Journal of Curriculum and Teaching*, 3(2), 2014.
- Rittle-Johnson, B., & Koedinger, K. R. “Comparing instructional strategies for integrating conceptual and procedural knowledge.” 2005.
- [Prodigy Education — 64% of Americans Struggle With Math Anxiety](#)
- [APA — Preventing Math Anxiety](#)